Aegis Futures Lab — User Manual

For traders and for anyone curious. Everything you need to use the app, in plain language.

Live app: https://aegis-futures-lab-khaki.vercel.app This manual is also inside the app on the **Guide** page (More → Guide).

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1. Start here

Aegis has a bot that watches two stock-index futures — the S&P micro (MES) and the Nasdaq micro (MNQ) — and tests trading methods on them. It shows you what it finds in plain words, and it keeps an honest score.

Aegis uses one **virtual $10,000 account**. Its **experimental learner** simulates trades on delayed prices and keeps an honest record of fills, costs and skips. **No method has beaten chance yet.** Virtual trading is practice, not proof that anything makes money. Trade ideas are not advice.

> **Nothing here touches real money.** There is no broker connection. Prices are delayed 10–15 minutes and trade ideas arrive 5–15 minutes after the setup. Use the app to practice, learn and keep score — never as a live trade instruction.

2. The five tabs

Tab	What it answers
Today	What the experimental learner is doing, what happened to its virtual money and what it last learned. Then the two markets, the latest trade ideas and any big news coming up.
Trades	Every idea the learner decided on, taken or skipped, with the reason. Tap one for the evidence: the values saved when it decided, the simulated fill, the exit, costs and the result.
Learn	The model in charge, the candidates being tested, every review's verdict and the full change history. Open Historical practice for the older-data study. Method research sits behind a separate panel; it does not show another account balance.
Chart	Prices, the price areas the bot watches, the learner's virtual trades as arrows, why there is no idea right now, and the big news it steps aside for.
More	Trade ideas (Ideas) and their history (Review), method research, historical practice, your Journal, this Guide, settings (ET or IST, hide money) and the research room.

Any word with a dotted underline can be tapped for a short meaning. Close a sheet with its × or Escape.

Read the widgets

- **Your day at a glance.** Today shows the bot's recorded activity, trades closed today, open trades and the next expected check. Tap a count to open Trades. Each market's Open chart link selects that market.
- **Account journey.** The line joins the starting balance, saved daily closes and the latest estimate. It is not a tick-by-tick chart. Open results can still change.
- **Trade cards.** The large number is the final result after costs, or an estimate for an open trade. The price bar shows the stop, actual entry and target. The dots below show whether the idea has filled and closed. Waiting and cancelled orders do not show a made-up result.
- **Find a trade.** Tap Open or Closed at the top of Trades, or use the filter row. Tap a card for the decision, fill and exit timeline, costs and saved evidence.
- **Decisions in this view.** The coloured strip counts closed, open, waiting, skipped and unfilled decisions in the cards you have loaded. It follows your filter. More cards can change these counts. They are not the account's lifetime totals.
- **Market clock and price range.** Chart shows time left in the entry window. Its ring follows that window, including early closes. The price range marks the last close between the low and high of the loaded candles. The dates below say which period it covers. It is not a forecast.
- **Find a tool.** In More, search for a tool by name or what it does. Clear the search to see all tools and the larger clock and privacy controls. These display settings stay on your device.

- **Open risk.** The bar shows how much of the account's risk allowance is in use. It is not a forecast of the next loss.
- **Learning progress.** On Learn, the ring counts usable examples toward the 50 needed to start training. It is not a score of skill or profitability. A fair review and fresh confirmation need more evidence.
- **More detail, when needed.** Tap Doing, Learning or Data for its reason. Open the timing details for the last check and next expected check. Your phone's reduced-motion setting turns off the animations. Hide money also hides the account chart.

3. How to read a trade idea

1. **Buy or Sell, and which market.** "Sell · Nasdaq micro" means the idea expected the Nasdaq micro to fall.
2. **The badge.** A red "Method hasn't beaten chance" means the method behind it was tested on years of prices and did no better than random entries. Amber "Not tested yet" means nobody has checked.
3. **The bar.** Stop on the left, target on the right, the white tick is the entry. The red part is the room to be wrong; the green part is how far it aimed. A ring shows where it ended.
4. **The sentence.** "Exits at X if wrong (−$Y per contract) or Z if right (+$W)" turns the prices into dollars for one contract. MES moves $5 a point, MNQ $2 a point.
5. **The ending.** Reached its target, Stopped out, Closed before stop or target, Open now — or "Not resolved yet" when the bot has not recorded how an old idea finished. Results are after costs.
6. **Details.** Tap a card for why the bot took it and, under "How this kind of idea has done", the numbers with their sample sizes.

4. The experimental learner

Meet the bot on Learn

The animated bot shows the latest recorded activity: watching for ideas, following virtual trades, organising examples, testing versions, waiting or needing attention. Watching means it is between scheduled checks, not training continuously. Offline or when a job has not reported back, the animation stops and the reason stays visible. You can pause the animation. It also rests off screen and follows your phone's reduced-motion setting.

My learning notebook shows usable examples at the last recorded check, with its date, plus versions trained, changes adopted and fresh results. Taken and skipped ideas can both supply examples. A bigger count is more evidence, not proof of better trading. The next milestone explains what is missing. Open **The path to a better bot** for all the steps.

Working on its own shows expected check, notebook and review times in your selected clock. The jobs continue when you close the app. **Recent activity** shows recorded job times and outcomes. Open **Example quality at the last check** for usable examples, waiting outcomes and exclusions. Its date matters; it is a report from that check.

The bot uses stored outcomes and small models on the existing service. It makes no paid AI calls or new data purchases. Identical training inputs share one calculation within a review. Unchanged datasets are reused. The normal review, risk and fresh-evidence requirements still apply. Usage limits can defer optional work; the app's usage guard is not a provider billing meter.

The learner is the bot's own **virtual $10,000** account. After each complete market update, it processes ideas and price bars in market-time order. The simulation stays 30 minutes behind received prices so source checks can finish before it passes an idea. This planned wait is separate from feed delay. It decides after the idea's bar closes and fills on the next available bar. It only uses prices available at that simulated moment. Skipped ideas are followed too, so it cannot learn only from its own winners.

1. **Limits.** At most $100 at risk on one trade and $200 across open trades. A $400 daily loss stops it until the next session. A $2,000 drop from its best stops the campaign for good. It never raises risk after a loss, and a new campaign keeps the earlier losses on record.
2. **Honest fills.** An order fills on the next price after the decision, with costs on both sides. A bar that touches both stop and target counts as the stop. A gap through the stop fills at the worse price. If the next market period has no price, the order waits; the app never invents a price. It can fill fewer contracts if risk is higher at the opening price. Filled trades show the actual size. Waiting and cancelled orders are labelled separately.

3. **Nightly.** Finished trades go into a numbered learning record. Nothing is learned from a single loss.
4. **Weekly.** It may test up to 3 new model versions from a fixed list, side by side with the current one on the same fresh ideas. A new version takes over only after 150 results it never trained on, 20 fresh trading days, 60 fresh ideas and two passing reviews at least 6 days apart. It must clearly beat the current model, not trading at all and random picks, after costs. The old version is kept and comes back automatically if the new one does worse.
5. **How long a test lasts.** At most 3 candidates are tested at a time, each for up to 52 weeks, because fresh results come slowly — about one usable idea a day. A candidate is retired after 6 inconclusive reviews in a row once it has 150 results it never trained on, or after 52 weeks without a passing review. A new candidate takes the free place.
6. **Why 50 is not enough.** Training can start at 50 finished trades. A fair review splits the record into 6 blocks by date, and each of its 5 test periods needs 50 earlier trades to learn from — roughly 300. Periods that are too thin are shown and skipped, never counted. Ideas too risky for one contract are not counted at all.

Three labels show its state, each with a reason. **Doing**: watching the market, managing trades, waiting, paused, stopped, last check failed or status unknown. **Learning**: collecting results, candidate ready, change adopted, candidate rejected or no proven gain yet. **Data**: current, late, stale or offline. A missing check reads "status unknown", never "running". "No proven gain yet" is an honest result, not a fault.

The repaired account replays ideas from 7 October 2026. Those trades and later delayed-market fills are labelled "Simulated on delayed prices". The time of the simulated decision and the actual time the idea was received are shown separately. Earlier skipped attempts remain under Trades → Earlier attempts. Replay results can help train and test candidates, but never count as fresh evidence for a model taking over. Synthetic test prices never count as market evidence. Only the owner can pause it through the project's GitHub controls. Offline, the app is read-only.

Behind the trades

Today, Trades and Learn show the day's source checks, eligible ideas, accepted ideas, filled trades, cancelled orders and learner skips. Accepted does not mean filled. Filled and cancelled counts follow decisions from the selected day, even if an order finishes later. Open an unfilled order for its reason. Choose a date in ET or IST. These are calendar-day counts; source setup checks below them follow the New York trading session.

Practice finished counts closed practice outcomes for those decisions. Ready for check counts results with the saved details needed for learning; the next learning check still applies its time cutoff and counts each idea once. Open Recorded decisions to follow each decision to its fill and learning status.

Account timing separates prices received from prices already simulated. A healthy feed can be ahead of the simulation while source checks finish. Truly late corrections are still skipped. Earlier fills, skips and balances are never rewritten to make activity look better.

Open a source method to see why its loss guard paused it and how its separate practice results are recovering. The guard pauses below a 0.80 profit-to-loss ratio over 20 usable closed results. It resumes after 15 in-pause results reach 1.10, or have wins with no losses, with at least three trading days between changes. A full progress bar alone does not pass the guard. Practice continues during a pause and never changes the learner's balance.

Open Paused method practice to see the selected day's recorded source simulations. Stale prices and uncertain fills are labelled. These results can change when the source price history changes; revised-away ideas stay visible but are not counted as results. They do not enter the learner, change its balance or count as a new study. If the loss guard cannot be checked, the source check stops and no new batch is published until a later check succeeds.

Open a skipped decision for its saved comparison: contract risk and remaining allowance, model odds and required odds, or the missing prerequisite. Older records may lack these values; the app says so. Decision time, receipt time, saved time and campaign context explain delayed processing. The check timeline separates normal waiting from missing prices, stale data and failed checks. A failed refresh shows a warning and keeps the last successful update for that day.

5. Historical practice

The bot also replays its own older market archive — the S&P and Nasdaq micros from 6 May 2019 to 23 September 2026 — to prepare a better candidate. It is practice on old data, kept apart from everything else.

1. **Only what it could know then.** At each old idea it decides with the prices it could have seen at that moment, using the historical study's original receipt-time rules, about 25 minutes late at the next 15-minute check. This study stays separate from the virtual account's delayed market clock. Gaps and contract-roll jumps in the old data are marked and the ideas near them left out — never patched.
2. **Counted once.** Each idea counts once, however many copies the records hold. Ideas too risky for one contract under the $100 limit are not counted.
3. **Rules first.** The dates, the 3 versions to try and every check were frozen before any old result was read. The oldest 60% of trading days trains, the next 20% checks, and the newest 20% is tested once. All 3 versions are kept, including the ones that fail.
4. **Fair comparisons.** Each version is compared on the same ideas with the current model, a simple average-rate guess, not trading at all and random picks — with costs doubled as a stress test and with the same account limits.
5. **Already seen.** Earlier research has looked at all of these dates, so the result is marked "already seen": practice, not proof.
6. **Then fresh ideas.** A candidate that is not rejected is frozen and watches new ideas beside the current model without changing a single decision. It must then pass every adoption check on fresh ideas — 20 fresh trading days, 60 fresh ideas and two passing reviews — like any other candidate. Old data counts for none of these. Practice results still go into the learning record. Fresh confirmation waits while the account uses delayed simulations. The older candidate remains saved.

Learn → Historical practice shows an evidence checklist: **Data ready**, **Historical test complete**, **Fresh confirmation waiting** and **Validated paper improvement**. There is no maturity score, and the last line never means proven for real money. "No validated improvement yet" is an honest result, not a fault.

The study never refills or resets the virtual account and never adds old profit to it. It runs at $0 on data the project already holds, in short capped runs, and pauses itself when the free limits run low.

6. One virtual account and separate records

The experimental learner uses the only active virtual account. Today, Trades and Learn show the same balance and trade history.

Trade ideas are the methods' separate research record. Their own results are never added to the account balance.

Your journal records your own decisions. It does not change the bot's money.

The earlier qualification account is retired. Its old record is kept for audit, and the qualification tests remain under More → Method research. Historical practice on years of older data stays separate and never adds old profit to the account.

7. How a method earns trust

1. **Watch.** Every 15 minutes, all futures week, the bot reads prices and posts any idea its methods spot. New ideas only start between 02:00 and 15:25 New York time (11:30–00:55 IST while the US is on summer time, 12:30–01:55 IST in winter), and everything is closed by 15:25 New York time.
2. **Test.** A method must beat thousands of random entries on years of past prices, pass a separate confirmation period, then earn 60 new trades over 20 trading days and two weekly reviews. Past replays never count as new trades.
3. **Qualification.** Passing every check records a research result. It does not start a second account or turn replay results into fresh proof. The virtual account keeps the limits listed above.

Every week the bot also tests any method marked "Not tested yet" against random entries by itself, and reports the result on Learn. It never promotes a method on its own — a person has to.

Open Learn → Method research → Weekly method screens to see recorded comparisons with random entries and with doing nothing ($0). Historical results are after modelled costs and show how many trades were tested, the price source and the dates covered. A positive historical total is not proof that a method will make money or has passed qualification.

Too little evidence is shown in amber, never red: too little data is not a loss. Red is only for a measured loss. Every rate is shown with its n, and below 30 trades it is marked "previewed, not judged".

8. Is the bot working?

Word	Meaning
Running	Price checks are arriving every 15 minutes.
Resting	The market is closed (weekends, holidays). That is the schedule, not a fault.
Prices delayed / Running late	The feed or a check is behind. Ideas catch up on the next pass.
Needs attention	Part of a check had a problem, such as the virtual account update. Trade ideas are still checked.
Last check failed	The newest check errored. The next one runs within 15 minutes; a watchdog alerts if it keeps failing.

The bell in the header lists anything worth a look. Its dot is amber for a warning and red only when something failed. News notes alone do not light it.

9. Your journal

More → Journal is yours alone. Pick the trading day, type in a trade or import your broker's file (Tradovate and Topstep exports work), and export it any time. Entry and exit times are typed in New York time to match the chart. Sign in with your email to keep a private copy across devices; local saving always works.

"Compare with the bot" re-runs the zone method over 60 days in your browser and shows what it did on your days. It takes a few seconds, so it only loads when you open it.

10. ET or IST

Times follow the ET/IST switch in the header (also under More → Settings). Phones in India start on IST. Two things stay in New York time on purpose: trading days (a day's ideas group by the New York date) and journal entry times. Session rules print both clocks in the app.

11. The research room

More → Research room holds the raw tools: the Strategy Lab (test a method on past prices yourself), Diagnostics (the beat-random test, market-year by market-year) and Data (import price files, replay a past day). They use statistics words on purpose; tap any underlined word, or see the list below.

Methods carry their standing everywhere: **red "hasn't beaten chance"** means tested and failed; **amber "not tested yet"** means unknown. Lab results never become trade ideas.

12. Put it on your phone

iPhone: open the site in Safari, tap Share, then Add to Home Screen. Android: open it in Chrome, tap ⋮, then Add to Home screen. It opens straight onto Today.

13. Words you'll see

Word	Meaning
Paper trading	Practice trades with imaginary money. Every trade in this app is a paper trade. Nothing here touches real money.
Delayed prices	Prices here are 10 to 15 minutes old, and trade ideas arrive 5 to 15 minutes after the setup. Never use them as a live order.
Practice money	The earlier qualification account, now retired. Virtual trading uses the learner's single account; the qualification tests remain available.
Experimental learner	The bot's own virtual $10,000 account. It takes trade ideas by itself, records every take and skip, and learns under fixed rules. Its results are evidence being collected, not proof.
Virtual equity	What the learner's pretend account is worth now: $10,000 plus finished results plus an estimate for open trades. It is pretend money only.
Model version	The rule the learner uses to take or skip ideas. Version 1 takes every idea. Later versions are small models trained on the learner's own finished trades.
Candidate	A new model version tested side by side with the current one on the same fresh ideas. It takes over only after every check passes, twice.
Rolled back	A version that took over was put back to the previous one because it did worse or gave an unusable answer. A rolled-back version can never return.
Shadow result	What an idea would have done with one contract, whether the learner took it or not. It lets the learner learn from skipped ideas too.
Synthetic prices	Made-up test prices used to prove the software works. Results on them never count as evidence that anything makes money.
Historical practice	The bot replays older market data, deciding each idea only with what it could have known then. It can prepare a candidate, but never counts as fresh trading days and never changes the virtual account.
Already seen	A stretch of older data that earlier research has already looked at. A result on it is practice, not proof, because the methods were shaped while looking at it.
Trade idea	A buy or sell setup the bot spotted and then followed to its end on delayed prices. It is a simulated record, not a trade in the practice account.
MES · S&P micro	The Micro E-mini S&P 500 future. Each 1-point move is worth $5 per contract.
MNQ · Nasdaq micro	The Micro E-mini Nasdaq-100 future. Each 1-point move is worth $2 per contract.
Entry	The price where the idea gets in.
Stop	The price that proves the idea wrong. If price reaches it, the trade closes with a planned loss. Fast markets can fill a little beyond it.
Target	The price where the idea takes its profit.
Reward to risk	How far the target is compared with the stop. 1.5 : 1 means the possible gain is one and a half times the possible loss.
Zone	A price area where strong buying or selling showed up before. Buy areas sit below price, sell areas above it.
Fresh zone	A zone price has not come back to yet. Tested means it has been touched once already.
Flat by 15:25 ET	Every idea closes by 15:25 New York time, just before the session ends. No overnight positions.
News pause	The bot takes no new ideas from 30 minutes before to 30 minutes after big US news like CPI, jobs or a Fed decision.
Win rate	The share of closed trades that made money. A high win rate can still lose money if the losses are bigger than the wins.
Profit factor	Money won divided by money lost. Above 1 means winners outweigh losers; below 1 means the method lost money.

Word	Meaning
Expectancy	The average result per trade after costs. Below zero means each trade loses money on average.
n (sample size)	How many trades a number is based on. Under 30 is too few to judge, so those numbers are shown faded and marked amber.
Likely range	Where the true value probably sits, given how few trades there are. A wide range means the number could easily be luck.
Costs	Commission ($1.20 per side per contract) and one tick of slippage on the way in and out. Every result here is after costs.
Slippage	Getting filled at a slightly worse price than planned. The app charges it on every trade.
Equity	What the practice account is worth right now, including any open trade.
Open risk	How much the practice account could lose if every open trade hit its stop.
Drawdown	How far the account has fallen from its highest point. A $2,000 drawdown locks the practice account until it is reset.
Beat-random test	The method's trades are compared with thousands of random entries on the same days. If it cannot beat random entries, it has no real edge.
Hasn't beaten chance	The method was tested on years of data and did no better than random entries. Its ideas are kept as a record, not as advice.
Not tested yet	Nobody has run the beat-random test on this method yet. Untested is not the same as losing — it just isn't known.
New trades	Trades the bot recorded live, as the prices arrived. Past replays never count, so a method has to prove itself going forward.
Qualify	A method earns qualification with a passed history test, a separate confirmation, 60 new trades over 20 trading days, and two weekly reviews. Delayed simulations never count as fresh proof.
Probation	The earlier qualification account used half risk ($50 per trade) for newly qualified methods. That account is now retired; the single virtual account follows its own fixed limits.
Zone setup and daily flow	The two kinds of trade ideas. Zone setups are rare returns to strong price areas. Daily flow fades short sharp moves, at most twice a day.
Paused by the breaker	When a kind of idea loses too often lately, the bot keeps simulating it but hides it from the results until it recovers.
Journal	Your own trades, typed in or imported from your broker. They stay separate from the bot and are private to you.

14. If something looks wrong

- A warning with "Try again" means a read failed; the screen keeps the last good numbers.
- An idea marked "Not resolved yet" is left out of every total until the bot records its end.
- A dash (—) means there is no number yet, not zero.
- An open learner trade marked "~" or "stale" has no fresh price; its result is an estimate.
- Still stuck? Check the bell, then Learn → health. The research drawers on Ideas show every excluded row.

Manual version: matches the app as of 2026-10-30 (paused practice and method screens). If the app has changed since, the in-app Guide is the source of truth.